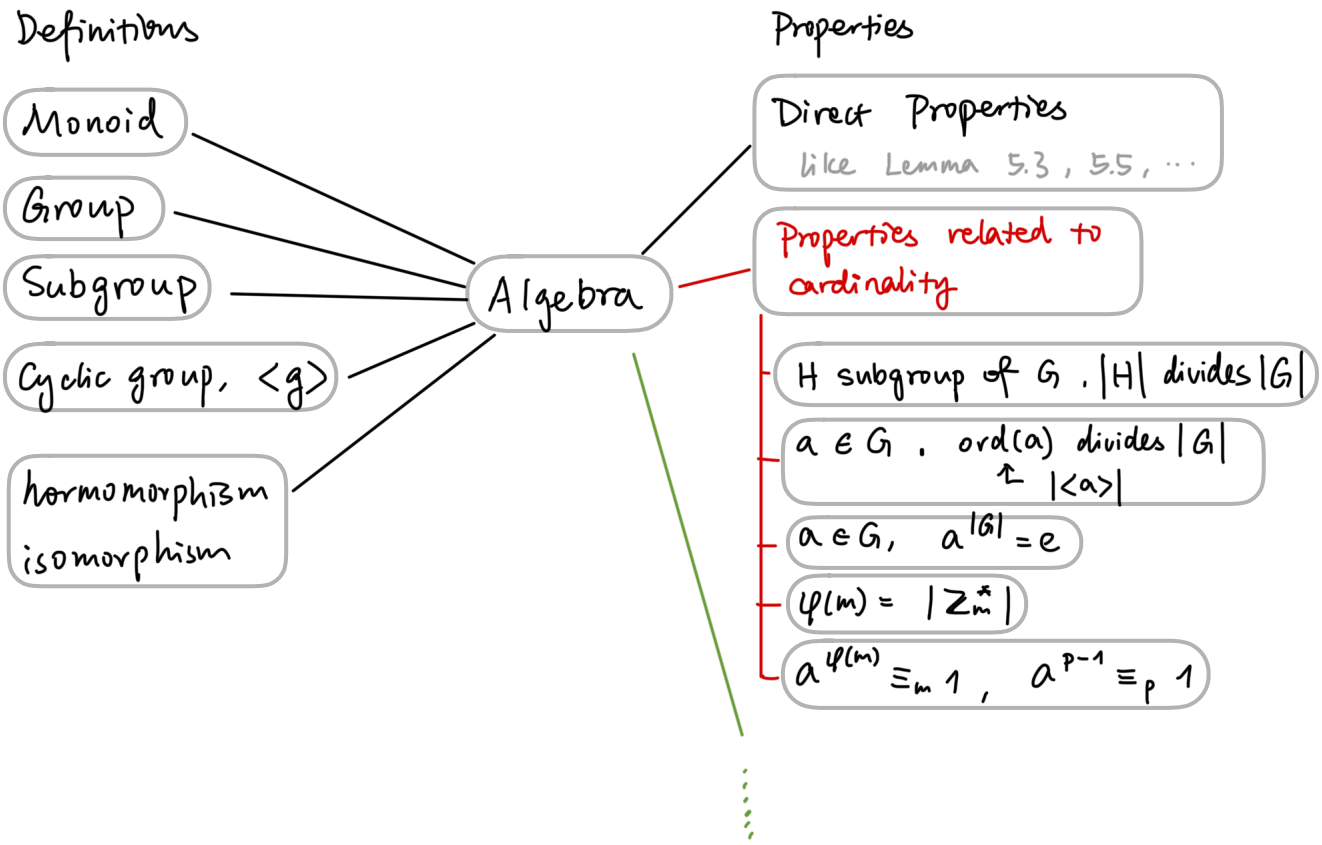
Week 8


Definitions
Monoid
Group
Subgroup
Cyclic group, <g>
homomorphism
isomorphism
Algebra
Properties
Direct Properties
like Lemma 5.3, 5.5, ···
Properties related to cardinality
H subgroup of G. |H| divides |G|
a ∈ G. ord(a) divides |G|
↑ |<a>|
a ∈ G, a^|G| = e
φ(m) = |Z*_m|
a^φ(m) ≡_m 1, a^(p-1) ≡_p 1

Monoids, Groups

Definition 5.1. An *operation* on a set S is a function[1] $S^n \to S$, where $n \geq 0$ is called the "*arity*"[2] of the operation.

Definition 5.2. An *algebra* (or *algebraic structure* or Ω-*algebra*) is a pair $\langle S; \Omega \rangle$ where S is a set (the *carrier*[3] of the algebra) and $\Omega = (\omega_1, \ldots, \omega_n)$ is a list of operations on S.[4]

$\langle \mathbb{Z}; +, -, 0, \cdot, 1 \rangle$

Neutral Element

Definition 5.3. A *left [right] neutral element* (or *identity element*) of an algebra $\langle S; * \rangle$ is an element $e \in S$ such that $e * a = a$ $[a * e = a]$ for all $a \in S$. If $e * a = a * e = a$ for all $a \in S$, then e is simply called *neutral element*.

Lemma 5.1. *If $\langle S; * \rangle$ has both a left and a right neutral element, then they are equal. In particular $\langle S; * \rangle$ can have at most one neutral element.*

Also possible: only left NE, no right NE.

Inverse Element

Definition 5.6. A *left [right] inverse element*[5] of an element a in an algebra $\langle S; *, e \rangle$ with neutral element e is an element $b \in S$ such that $b * a = e$ $[a * b = e]$. If $b * a = a * b = e$, then b is simply called an *inverse* of a.

Lemma 5.2. *In a monoid $\langle M; *, e \rangle$, if $a \in M$ has a left and a right inverse, then they are equal. In particular, a has at most one inverse.*

Monoid

Definition 5.5. A *monoid* is an algebra $\langle M; *, e \rangle$ where $*$ is associative and e is the neutral element.

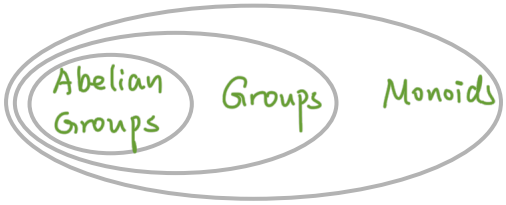
Group

Definition 5.7. A *group* is an algebra $\langle G; *, \widehat{\ }, e \rangle$ satisfying the following axioms:

- **G1** $*$ is associative.
- **G2** e is a neutral element: $a * e = e * a = a$ for all $a \in G$.
- **G3** Every $a \in G$ has an inverse element $\widehat{a}$, i.e., $a * \widehat{a} = \widehat{a} * a = e$.

} monoid

Abelian Group: $*$ is also commutative



	Monoid?	Group?	$*$ commutative?
$\langle \mathbb{Z}; + \rangle$	✓ 0 = NE	✓ $a = -a$	✓ + comm.
$\langle \mathbb{Z}; \cdot \rangle$	✓ 1 = NE	✗ $2 \cdot \frac{1}{2} \notin \mathbb{Z}$	✓ ✗ abelian group
$\langle \mathbb{R}; \cdot \rangle$	✓ 1 = NE	✗ 0 has no inv.	✓
$\langle \mathbb{Z}; - \rangle$	✗ $(a-b)-c \neq a-(b-c)$ $= a-b+c$	✗	✗

Lemma 5.3. *For a group $\langle G; *, \widehat{\ }, e\rangle$, we have for all $a, b, c \in G$:*

(i) $\widehat{(\widehat{a})} = a.$

(ii) $\widehat{a * b} = \widehat{b} * \widehat{a}.$

(iii) *Left cancellation law:* $a * b = a * c \implies b = c.$

(iv) *Right cancellation law:* $b * a = c * a \implies b = c.$

(v) *The equation $a * x = b$ has a unique solution x for any a and b. So does the equation $x * a = b$.*

$a^{-1} * a * x = a^{-1} * b$

$e * x = a^{-1} * b$

$x = a^{-1} * b$

Definition 5.9. The *direct product* of n groups $\langle G_1; *_1\rangle, \ldots, \langle G_n; *_n\rangle$ is the algebra

$$\langle G_1 \times \cdots \times G_n; \star\rangle,$$

where the operation $\star$ is component-wise:

$(e_1, e_2 \ldots e_n)$

$$(a_1, \ldots, a_n) \star (b_1, \ldots, b_n) = (a_1 *_1 b_1, \ldots, a_n *_n b_n).$$

Lemma 5.4. *$\langle G_1 \times \cdots \times G_n; \star\rangle$ is a group, where the neutral element and the inversion operation are component-wise in the respective groups.*

Example: $\langle \mathbb{Z}_5; \oplus\rangle \times \langle \mathbb{Z}_7; \oplus\rangle = \langle \mathbb{Z}_5 \times \mathbb{Z}_7; *\rangle = G$

is again a group

G1 (Associative) $((a,b) * (c,d)) * (m,n) = (R_5(a+c), R_7(b+d)) * (m,n)$

G2 (NE) $\forall (a,b) \in \mathbb{Z}_5 \times \mathbb{Z}_7$

$(0,0) * (a,b) = (R_5(0+a), R_7(0+b)) = (R_5(R_5(a+c)+m), R_7(R_7(b+d)+n))$

$= (R_5(a), R_7(b)) = (R_5(a+c+m), R_7(b+d+n))$

$= (a,b) = (R_5(a+R_5(c+m)), R_7(b+R_7(d+n)))$

G3 (Inverse) $\forall (a,b) \in \mathbb{Z}_5 \times \mathbb{Z}_7 = (a,b) * (R_5(c+m), R_7(d+n))$

$\exists\ a^{-1} \in \mathbb{Z}_5 \quad \exists\ b^{-1} \in \mathbb{Z}_7 = (a,b) * ((c,d) * (m,n))$

s.t. $a * a^{-1} = e \quad \wedge \quad b \cdot b^{-1} = e$

$(a,b) * (a^{-1}, b^{-1}) = \ldots$ $\quad G2' \wedge G3' \Rightarrow G3 \wedge G2$

5.2.4 (Non-)minimality of the Group Axioms

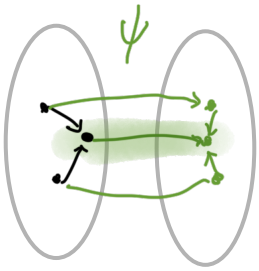
In mathematics, one generally wants the axioms of a theory to be minimal. One can show that the group axioms as stated are not minimal. One can simplify axiom **G2** by only requesting that $a * e = a$; call this new axiom **G2′**. The equation $e * a = a$ is then implied (by all axioms). The proof of this fact is left as an exercise. Similarly, one can simplify **G3** by only requesting that $a * \widehat{a} = e$; call this new axiom **G3′**. The equation $\widehat{a} * a = e$ is then implied. The proof for this is as follows:

Homomorphism & Isomorphism

Definition 5.10. For two groups $\langle G; *, \widehat{\ }, e\rangle$ and $\langle H; \star, \widetilde{\ }, e'\rangle$, a function $\psi : G \to H$ is called a *group homomorphism* if, for all a and b,

$$\psi(a * b) = \psi(a) \star \psi(b).$$

If ψ is a bijection from G to H, then it is called an *isomorphism*, and we say that G and H are isomorphic and write $G \simeq H$.



Lemma 5.5. *A group homomorphism ψ from $\langle G; *, \widehat{\ }, e\rangle$ to $\langle H; \star, \widetilde{\ }, e'\rangle$ satisfies*

(i) $\psi(e) = e'$,

(ii) $\psi(\widehat{a}) = \widetilde{\psi(a)}$ *for all* a.

$\langle \mathbb{Z}_m \times \mathbb{Z}_n, * \rangle$

Example: $\langle \mathbb{Z}_m ; + \rangle \times \langle \mathbb{Z}_n ; + \rangle \simeq \langle \mathbb{Z}_{mn} ; +_{mn} \rangle$ iff $\gcd(m,n) = 1$.

prove isom.

Homomorphism: $\psi(a * b) = \psi(a) * \psi(b)$

$\psi : \mathbb{Z}_{mn} \to \mathbb{Z}_m \times \mathbb{Z}_n \quad , \quad x \mapsto (R_m(x), R_n(x))$

$$\begin{aligned} \psi(a +_{mn} b) &= \left(R_m(a +_{mn} b) , R_n(a +_{mn} b) \right) && | \text{ def } \psi \\ &= \left(R_m\left(R_m(a) + R_m(b)\right) , R_n\left(R_n(a) + R_n(b)\right) \right) \\ &= \left(R_m(a), R_n(a) \right) * \left(R_m(b), R_n(b) \right) && | \text{ def } * \\ &= \psi(a) * \psi(b) && | \text{ def } \psi \end{aligned}$$

$R_m(a +_{mn} b)$
$= R_m(a + b - q\,mn)$
$= R_m(a + b)$
$= R_m(R_m(a) + R_m(b))$

ψ bijective: $\Leftrightarrow \exists\ \psi^{-1} : \mathbb{Z}_m \times \mathbb{Z}_n \to \mathbb{Z}_{mn}$

$\psi^{-1}((a,b)) = x$ with $\begin{cases} x \equiv_m a \\ x \equiv_n b \end{cases}$ unique $x \in \mathbb{Z}_{mn}$

$\psi^{-1}(\psi(x)) = x'$ $\quad \begin{cases} x' \equiv_m a \\ x' \equiv_n b \end{cases}$

$= x$

surj: $\forall\ (a,b) \in \mathbb{Z}_m \times \mathbb{Z}_n$

$\exists\ x \quad \psi(x) = (a,b) \qquad x = \psi^{-1}((a,b))$

inj: $\psi(x) = \psi(x')$

x and x' both solve $\begin{cases} x \equiv_m a \\ x \equiv_n b \end{cases} \wedge x, x' \in \mathbb{Z}_{mn}$

$x = x'$ according to CRT

Subgroups, cyclic groups, generators

Definition 5.12. Let G be a group and let a be an element of G. The *order*[8] of a, denoted $\mathrm{ord}(a)$, is the least $m \geq 1$ such that $a^m = e$, if such an m exists, and $\mathrm{ord}(a)$ is said to be infinite otherwise, written $\mathrm{ord}(a) = \infty$.

can be infinite

By definition, $\mathrm{ord}(e) = 1$. If $\mathrm{ord}(a) = 2$ for some a, then $a^{-1} = a$; such an a is called self-inverse.

$a^2 = e$

$a \cdot a = e$

Lemma 5.6. *In a finite group G, every element has a finite order.*

Definition 5.13. For a finite group G, $|G|$ is called the *order* of G.[9]

always finite

Subgroups:

Definition 5.11. A subset $H \subseteq G$ of a group $\langle G; *, \widehat{\ }, e\rangle$ is called a *subgroup* of G if $\langle H; *, \widehat{\ }, e\rangle$ is a group, i.e., if H is closed with respect to all operations:

(1) $a * b \in H$ for all $a, b \in H$, — Closure

(2) $e \in H$, and — NE

(3) $\widehat{a} \in H$ for all $a \in H$. — inv.

Trivial subgroup: $\{e\}$, G

Theorem 5.8 (Lagrange). *Let G be a finite group and let H be a subgroup of G. Then the order of H divides the order of G, i.e., $|H|$ divides $|G|$.*

Example: Find all subgroups of $\langle \mathbb{Z}_2; + \rangle \times \langle \mathbb{Z}_2; + \rangle$.

$|G| = 4$

① $\{e\}$, G: $\{e\}$ $\{(0,0), (0,1), (1,0), (1,1)\}$

$|H| = 1, 2, 4$

② subgroups of order 2:

H subgroup G

$\{(0,0), (0,1)\}$ ✓

$\{(0,0), (1,0)\}$ ✓

$\{(0,0), (1,1)\}$ ✓

Cyclic groups:

Definition 5.14. For a group G and $a \in G$, the *group generated by a*, denoted $\langle a \rangle$, is defined as

$$\langle a \rangle \stackrel{\text{def}}{=} \{a^n \mid n \in \mathbb{Z}\}. = \{(a^{-1})^n \mid n \in \mathbb{Z}\}$$

Example: $\langle \mathbb{Z}_{12}; \oplus \rangle$

Subgroups:

1: $\{0\}$

2: $\{0, 6\}$

3: $\{0, 4, 8\}$

4: $\{0, 3, 6, 9\}$

6: $\{0, 2, 4, 6, 8, 10\}$

12: $\mathbb{Z}_{12}$

Definition 5.15. A group $G = \langle g \rangle$ generated by an element $g \in G$ is called *cyclic*, and g is called a *generator* of G.

Being cyclic is a special property of a group. Not all groups are cyclic! A cyclic group can have many generators. In particular, if g is a generator, then so is g^{-1}.

Example 5.27. The group $\langle \mathbb{Z}_n; \oplus \rangle$ is cyclic for every n, where 1 is a generator. The generators of $\langle \mathbb{Z}_n; \oplus \rangle$ are all $g \in \mathbb{Z}_n$ for which $\gcd(g, n) = 1$, as the reader can prove as an exercise.

Example 5.28. The additive group of the integers, $\langle \mathbb{Z}; +, -, 0\rangle$, is an infinite cyclic group generated by 1. The only other generator is -1.

Corollary 5.10. *Let G be a finite group. Then $a^{|G|} = e$ for every $a \in G$.*

$\langle a \rangle$ subgroup of G

Corollary 5.11. *Every group of prime order*[10] *is cyclic, and in such a group every element except the neutral element is a generator.*

$k = |\langle a \rangle|$ divides $|G|$

$a^k = e$ $a^{|G|} = e$

$|G|$ prime p 1 p

$\langle a \rangle = |G|$ if $\mathrm{ord}(a) = p$

Corollary 5.9. *For a finite group G, the order of every elements divides the group order, i.e., $\mathrm{ord}(a)$ divides $|G|$ for every $a \in G$.*

It is easy to see that $\langle a \rangle$ is a group, actually the smallest subgroup of a group G containing the element $a \in G$. For finite groups we have

$$\langle a \rangle \stackrel{\text{def}}{=} \{e, a, a^2, \ldots, a^{\mathrm{ord}(a)-1}\}.$$

$\mathbb{Z}_m^*$ and Euler's function

Definition 5.16. $\mathbb{Z}_m^* \stackrel{\text{def}}{=} \{a \in \mathbb{Z}_m \mid \gcd(a, m) = 1\}$.

Theorem 5.13. *$\langle \mathbb{Z}_m^*; \odot, ^{-1}, 1 \rangle$ is a group.*

Example: $\mathbb{Z}_{12}^* = \{1, 5, 7, 11\}$

Definition 5.17. The *Euler function* $\varphi : \mathbb{Z}^+ \to \mathbb{Z}^+$ is defined as the cardinality of $\mathbb{Z}_m^*$:

$$\varphi(m) = |\mathbb{Z}_m^*|.$$

Lemma 5.12. *If the prime factorization of m is $m = \prod_{i=1}^{r} p_i^{e_i}$, then*[11]

$$\varphi(m) = \prod_{i=1}^{r} (p_i - 1) p_i^{e_i - 1}$$

$12 = 4 \cdot 3 = 2^2 \cdot 3^1$

$\varphi(12) = 4$ $\qquad 1 \cdot 2^1 \cdot 2 \cdot 3^0 = 4$

Corollary 5.14 (Fermat, Euler). *For all $m \geq 2$ and all a with $\gcd(a, m) = 1$,*

$$a^{\varphi(m)} \equiv_m 1.$$

$a^{|G|} \equiv_m 1$ $\qquad \varphi(m) = |\mathbb{Z}_m^*| = |G|$

In particular, for every prime p and every a not divisible by p,

$$a^{p-1} \equiv_p 1.$$

$|\mathbb{Z}_m^*| = p - 1$ $\qquad 1 \ldots p-1$

Theorem 5.15. *The group $\mathbb{Z}_m^*$ is cyclic if and only if $m = 2$, $m = 4$, $m = p^e$, or $m = 2p^e$, where p is an odd prime and $e \geq 1$.*

$2 \qquad 2 \cdot 2^1$

Theorem 5.7. *A cyclic group of order n is isomorphic to $\langle \mathbb{Z}_n; \oplus \rangle$ (and hence abelian).*

If G is cyclic with $|G| = n$, then there is exactly one subgroup H for every divisor d of n, $|H| = d$ and H cyclic.

(Not in script) Intuition: G isom. $\mathbb{Z}_n$ — G behaves the same way as $\mathbb{Z}_n$.

Example: find all subgroups of $\mathbb{Z}_{11}^*$ (FS 15)

$11 = p$ $\qquad \mathbb{Z}_{11}^*$ cyclic (Thm. 5.15) $\qquad |\mathbb{Z}_{11}^*| = 10$

$\mathbb{Z}_{11}^*$ isom. $\mathbb{Z}_{10}$ $\qquad 1, 2, 5, 10$

2: $\{1, 10\}$

5: $\{1, 3, 4, 5, 9\}$

~~$\langle 2 \rangle = \mathbb{Z}_{11}^* = \{2, 4, 8, 5, 10, 9, 7, 3, 6, 1\}$~~

HS16

e) ($\star\star\star$) Wie viele Isomporphismen gibt es zwischen zwei zyklischen Gruppen der Ordnung $n > 0$? Begründen Sie ihre Antwort. *(6 Punkte)*

Hinweis: Eine zyklische Gruppe der Ordnung n hat $\varphi(n)$ Generatoren.

$\psi : G \to H$

$\psi(g) = h$

g gen. of G

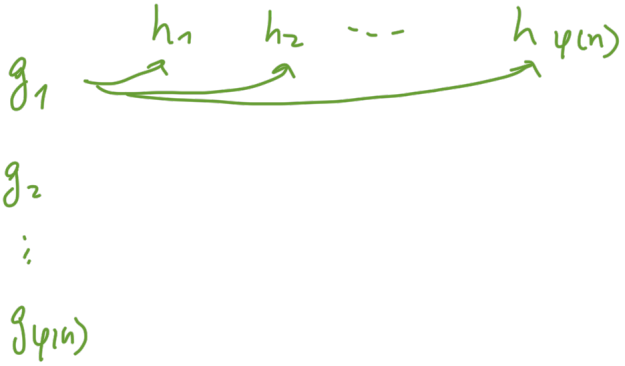
h gen. of h

G isom. $\mathbb{Z}_n$

$\forall g \quad g \in \mathbb{Z}_n^*$

$\varphi(n)$ generators,

$$\psi(g^k) = [\psi(g)]^k = h^k \qquad \forall\, k \in \mathbb{Z}$$



$\varphi(n)$ isom.

FS19

b) ($\star\star\star$) Let G be a *finite* non-empty set and let $*$ be a binary operation on G, such that:

- $*$ is associative and commutative,
- for all $a, b, c \in G$, if $a * b = a * c$ then $b = c$.

Prove that G is a group. That is, show (1) that there exists a (neutral element) $e \in G$, such that $a * e = e * a = a$ for all $a \in G$, and (2) that for every $a \in G$ there exists an (inverse) $b \in G$ such that $a * b = e$, where e is from (1).

Hint: Consider a sequence $a, a^2, a^3, \ldots$ for $a \in G$ and use the fact that G is finite. *(6 Points)*

$\exists k, l \in \mathbb{N} \quad a^k = a^l \qquad k < l \qquad \forall\, a \in G$

$a^{l-k} = e$

$a, a^2, \ldots$

$a^k = a^l$

$a \cdot a^{l-k-1} = e$

$l - k - 1 \geq 0$

$\forall\, b \in G \quad a^k \cdot a^{l-k} \cdot b$

$= a^l \cdot b$

$= a^k \cdot b$

$\Rightarrow$

$a^{l-k} \cdot b = b$

a^{l-k} is NE by G2'